CONTENTS

14.1 INTRODUCTION 2

14.2 ASSESSMENT METHODOLOGY 3
Scope 3
Data Sources 3
Assessment Approach 4
Significance criteria 4
Uncertainties and limitations 5

14.3 RELEVANT POLICY 6
Draft Swindon Borough Local Plan 2026: Submission Document (June 2013) 6

14.4 BASELINE CONDITIONS 7
Water 7
Electricity 7
Gas 8
Telecommunications 9

14.5 POTENTIAL EFFECTS 11
Water 11
Electricity 12
Gas 14
Telecommunications 15

14.6 MITIGATION MEASURES 18
Construction stage 18
Post-completion stage 18

14.7 RESIDUAL EFFECTS 20
Construction stage 20
Post-completion stage 20
Summary of effects 20

14.1 INTRODUCTION

14.1.1 This chapter has been prepared by Brookbanks Consultanting Ltd to present the findings of an assessment of the effects of the Proposed Development on service supply and utilities. The chapter also outlines a services supply strategy for the Proposed Development detailing the off-site works required to accommodate the additional supply loadings.

14.2 ASSESSMENT METHODOLOGY

Scope

14.2.1 The Study Area for the assessment covers the extent of the Site in relation to existing utilities and the potential effect the Proposed Development will have on any diversionary/protection works required.

14.2.2 In relation to the appraisal of the capacity of the existing utility networks surrounding the Site, the surrounding existing water, gas, electricity and telecommunication networks have been assessed to help determine the effects of the Proposed Development.

Data Sources

14.2.3 During the undertaking of this assessment, the following statutory bodies and interested parties have been consulted regarding the location of existing utilities all of which, where identified, have been transferred onto a composite service location plan (see **Appendix 14.1**).

- Thames Water
- Scottish and Southern Electricity
- Wales and West Utilities
- British Telecommunications
- Virgin Media
- Cable and Wireless
- Colt
- Instalcom

14.2.4 During the undertaking of this assessment, the following statutory bodies and interested parties have also been consulted regarding the available capacity:

- Thames Water (Water Supply)
- Scottish and Southern Electricity (Electricity Supply)
- Wales and West Utilites (Gas Supply)

- British Telecommunications (Telecoms Supply)

Assessment Approach

14.2.5 A survey has been completed to identify the existing utilities and features within the Site boundary.

14.2.6 Methods of assessment have been employed that are consistent with current guidance and recommendations in the form of statutory documents and recognised publications to ensure the findings are robust.

Significance criteria

14.2.7 Tables 14.1, 14.2 and 14.3 below outline the criteria for determining the magnitude of identified impacts, the sensitivity of receptors and the significance of the resulting effects.

Table 14.1: Magnitude

MAGNITUDE	CRITERIA
Large	Loss of supply for existing properties.
Moderate	Net reduction in supply for existing properties.
Small	Short term supply disruptions.
Negligible	Minor temporary change over a limited/local area

Table 14.2: Sensitivity

SENSITIVITY	RECEPTOR
High	Development located in an area with a lack of local utility capacity (electric, gas, potable water, foul water treatment or quality issues and telecoms). Therefore requiring significant offsite network reinforcements
Medium	Development located in an area where currently there are capacity issues on the existing utility networks to supply at least one of the utility services
Low	Development located in an area where currently there are minor capacity issues on the existing utility networks to supply the utility services
Negligible	Development located in an area where there is enough capacity to supply the development

Table 14.3: Significance

MAGNITUDE	SENSITIVITY			
	High	**Medium**	**Low**	**Negligible**
Large	Major	Major	Moderate	Minor
Moderate	Major	Moderate	Minor	Negligible
Small	Moderate	Minor	Minor	Negligible
Negligible	Minor	Negligible	Negligible	Negligible

Uncertainties and limitations

14.2.5 The analysis of the utilities and impacts is reflective on a point in time analysis of the utilities network, any alterations to the networks carried out by the relevant utilities providers cannot be anticipated nor taken account within this assessment.

14.3 RELEVANT POLICY

Draft Swindon Borough Local Plan 2026: Submission Document (June 2013)

14.3.1 Policy IN1 sets out a requirement for development to meet the cost of new infrastructure made necessary by the development, to include strategic, local and site specific infrastructure and to mitigate the impact of development on existing infrastructure.

14.3.2 Policy IN3 refers to the provision of infrastructure, and proposes a requirement for development to make provision to incorporate super-fast broadband where possible.

14.4 BASELINE CONDITIONS

Water

14.4.1 Potable water supplies are distributed by a network of water mains within the area of Swindon. The network is owned and operated by Thames Water. The nearest public water supply main is 600mm in diameter and runs about 2km north west of the Site.

14.4.2 The water supply network within the vicinity of the Site is fed from the Blunsdon Reservoir via the Blunsdon to Honda trunk main which is 600mm in diameter. The Blunsdon to Honda trunk main provides supplies to the Honda plant to the north west of the Site, as well as providing supplies to the urban area of Swindon to the west of the A419. Figure 14.4 shows the approximate location of the reservoir and the route of the trunk water supply main.

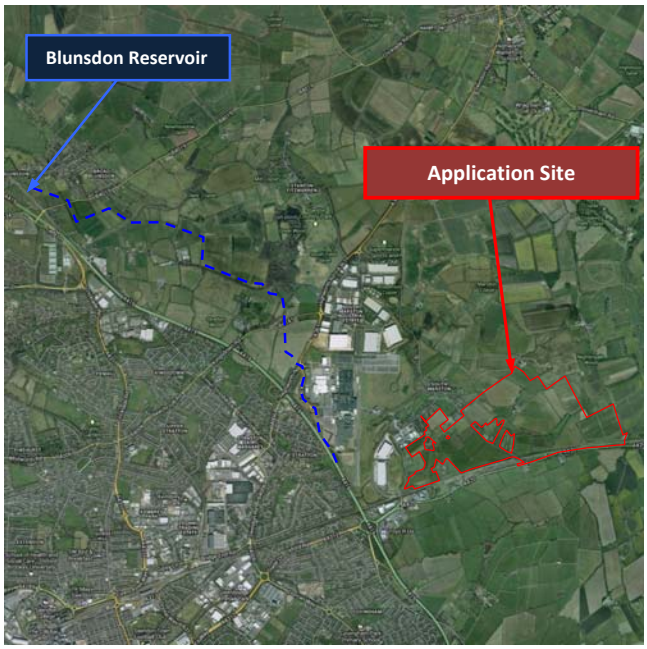

Figure 14.4: Location of Blunsdon Reservoir and route of 600mm Trunk Main

Electricity

14.4.3 The electricity assets within the area of Swindon are predominantly owned by Scottish and Southern Electricity (SSE) along with some third party suppliers.

14.4.4 Scottish and Southern Energy Power Distribution (SSE) has been consulted regarding the locations of existing network equipment in the vicinity of the Site. A network of mains distributes electricity to the surrounding residential and commercial areas.

14.4.5 Details of the existing apparatus are shown on a composite services plan, which is contained within **Appendix 14.1.**

14.4.6 Within the Site, 11kV overhead cables traverse in several locations before terminating at Stratton Primary Substation adjacent to Kingsdown Road, located approximately 2km to the north west of the Site.

14.4.7 The Stratton Primary Substation provides electricity supplies across the Site, as well as the surrounding areas. Figure 14.5 below shows the location of Stratton Primary Substation in relation to the Site.

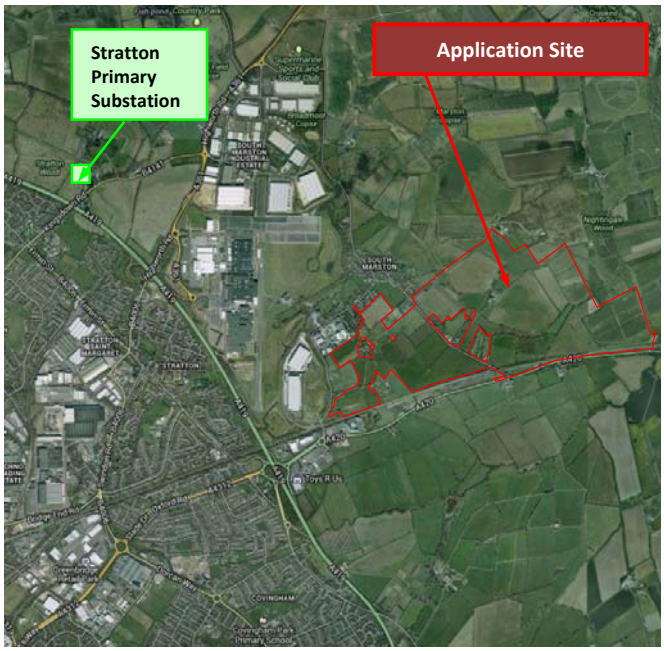

Figure 14.5.: Location of Stratton Primary Substation

Gas

14.4.8 Wales and West Utilities (WWU) has provided details relating to its existing network. The plans provided indicate a network of low and medium pressure gas mains within the surrounding residential areas.

14.4.9 WWU has also confirmed that there are no gas mains located within the Site and as such it is not anticipated that any diversionary works will be required. However, the existing gas supply network within the highways infrastructure in the vicinity of the Site may require some protection works, subject to the final site access proposals.

14.4.10 The nearest relevant main to the Site is a 90mm medium pressure main situated within Thornhill Road, adjacent to the western boundary of the Site.

14.4.11 The composite service location plan within **Appendix 14.1** contains details relating to the existing Wales and West Utilities apparatus.

Telecommunications

14.4.12 BT has provided details relating to its existing network. The plans provided indicate a telecommunication network within the surrounding residential areas to the north of the Site.

14.4.13 The principal routes for telecommunication cables within the area are Thornhill Road, Old Vicarage Lane and the A420. Overhead cables are also present in the residential areas around the Site that link to existing properties.

14.4.14 No BT apparatus is present within the boundary of the site.

14.4.15 Existing BT apparatus locations are detailed in the composite service location plan included within **Appendix 14.1.**

14.4.16 A number of other telecommunication services have been identified within the vicinity of the Site but not within the boundary. These include services operated by the following providers:

- Virgin Media
- Cable & Wireless
- Colt
- Instalcom.

14.4.17 The locations of the apparatus mentioned above have been detailed in the composite services plan included within **Appendix 14.1**.

14.4.18 The following utilities companies were consulted but have confirmed that they did not have plant within the vicinity of the Site and as such will not be affected by the Proposed Development:

- Level 3 Communications
- Orange
- VTESSE Networks
- Verizon Business
- Energetics
- Gamma Telecom
- GTC
- Sky Network Services
- Fibre Span Ltd
- Envoy
- Trafficmaster

14.5 POTENTIAL EFFECTS

Water

14.5.1 Water supply assessments have been completed to predict a flow of up to 12 l/s being required for the Proposed Development.

14.5.2 Water supply load increments have been estimated to give the following forecast loading profile:

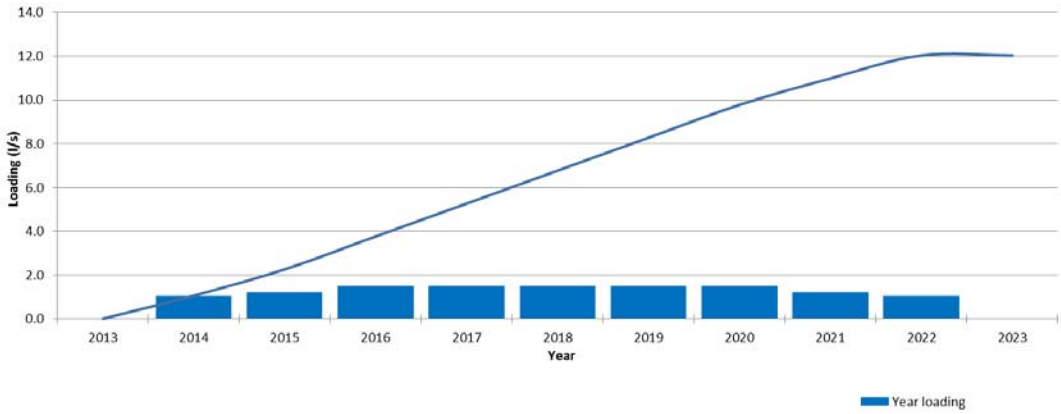

Figure 14.6: Predicted water supply loading profile

14.5.3 The above assessment assumes the completion of the Proposed Development in 2022, which is a worse case than the 2026 completion date assumed more widely in this EIA, thereby presenting a robust basis for the capacity assessments being undertaken by the supply company. This also applies to the other services under consideration in this chapter.

14.5.4 Discussions with Thames Water confirm the water supply trunk mains along Kingsdown Road are likely to be the preferred point of supply for the Proposed Development.

14.5.5 It is anticipated that the Proposed Development will connect to the existing 600mm water supply main within Kingsdown Road distributing supplies from this location to the proposed on-site buildings. The exact location of the point of connection will be confirmed at detailed design stage and through further hydraulic modelling which is currently being undertaken by Thames Water.

14.5.6 Figure 14.7 shows the 600mm main within Kingsdown Road and indicative route to the Site.

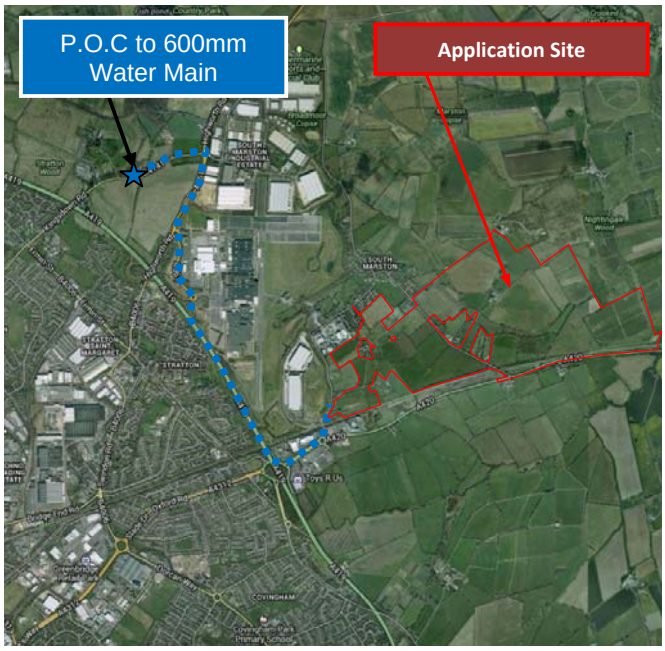

Figure 14.7: Indicative Water Supply Route to Site

Electricity

14.5.7 Electricity loading assessments have been completed and predict a total load of around 4,000 kWh being required for the Proposed Development.

14.5.8 Electricity load increments have been estimated to give the following forecast loading profile.

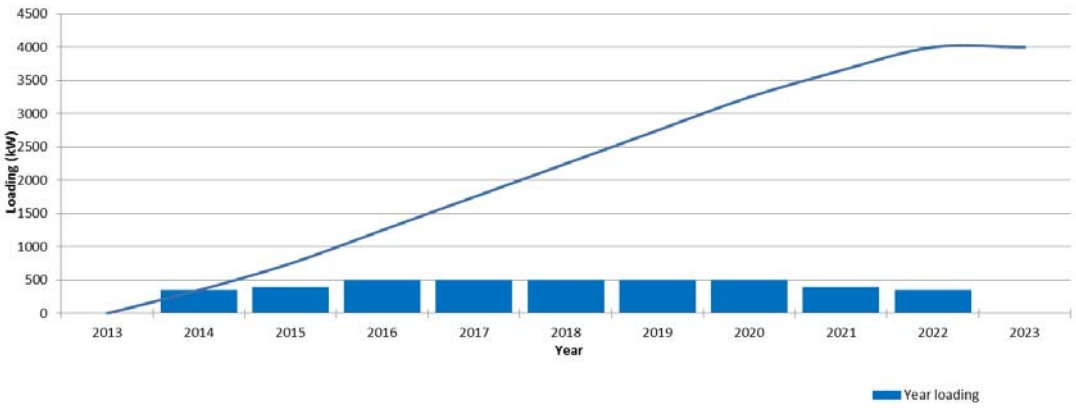

Figure 14.8: Predicted electricity supply loading profile

14.5.9 Discussions with SSE have confirmed a point of connection will be taken from the existing 132kV Stratton Primary Substation, along Kingsdown Road, with a new 11kV route to the Site.

14.5.10 The anticipated reinforcements which have been confirmed by SSE include a new 32kV-11kV transformer at the Primary Substation.

14.5.11 It is also anticipated that the overhead cables will be diverted underground by re-routing them through the on-site highway infrastructure. However, further investigations, in the form of a developer funded feasibility study, are required to ascertain reinforcement details and associated costs.

14.5.12 Figure 14.9 shows the anticipated route to bring a supply to the Proposed Development.

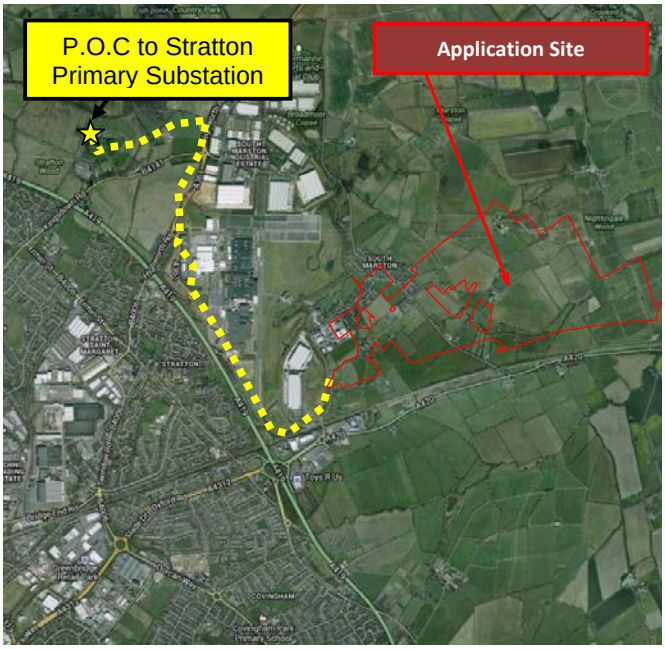

Figure 14.9: Indicative Electricity Supply Route to Site

Gas

14.5.13 Gas loading assessments have been completed and predict a total load of 4,200m³/hr being required for the Proposed Development.

14.5.14 Gas load increments have been estimated to give the following forecast loading profile.

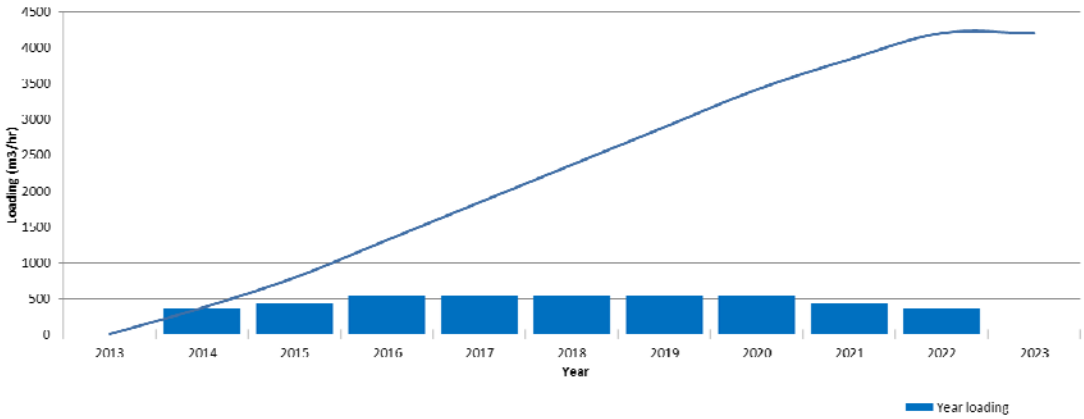

Figure 14.10: Predicted gas supply loading profile

14.5.15 Discussions with Wales and West confirm the 90mm medium pressure gas main along Thornhill Road is to be the likely point of connection.

14.5.16 It is anticipated that supplies will be distributed via a district medium to low pressure governor station. A governor for a site of this size would typically require an area of 4mx3m above ground, although they can also be housed in an underground manhole or the like to reduce the area of land required.

14.5.17 Figure 14.11 shows the 90mm main within Thornhill Road and indicative route to Site.

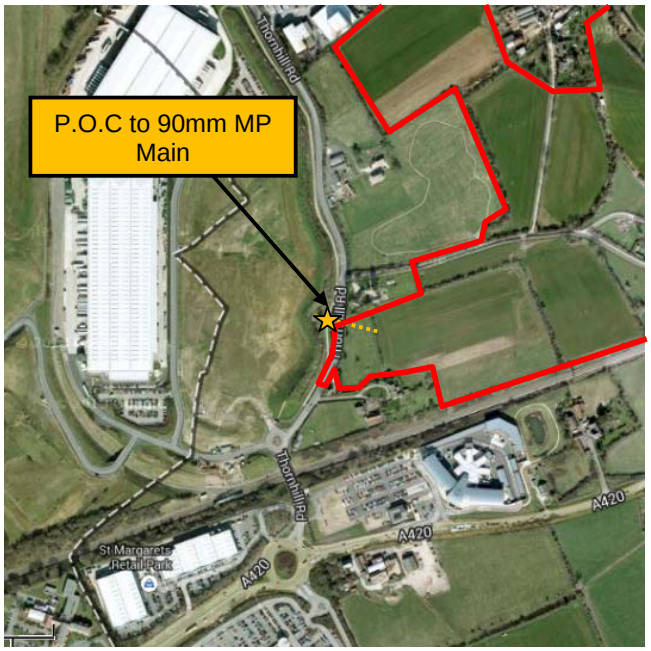

Figure 14.11: Indicative Gas Supply Route to Site

Telecommunications

14.5.18 Telephone connections can be made via the adjacent network facilities in and around the Site. Enquiries confirm that broadband ADSL services are available at the local BT Stratton St Margaret Exchange with download speeds up to 1.5Mbs.

14.5.19 The exchange currently serves approximately 7,859 residential properties and 279 non-residential premises.

14.5.20 Currently the communication providers (CP) that are available at the Stratton St Margaret Exchange, other than BT, are AOL, 02, Cable and Wireless, Sky and Talk Talk.

14.5.21 Further investigations also confirm that Fibre to the Cabinet (FTTC) is now available within the exchange area. This enables customers to experience download speeds of up to 40Mbps and upstream speeds of up to 10Mbps.

14.5.22 Virgin Media have confirmed the existence of plant in the area which will provide an additional option for supplying the Site with high-speed broadband.

14.5.23 In addition to BT and Virgin Media, there is also the potential for a fibre network to be implemented onsite through an alternative network operator, such as Independent Fibre Networks Limited (IFNL). This will provide faster connection speeds than what is available via a standard BT connection. Independent Fibre Networks (IFNL) are currently installing fibre on a number of smaller sites across the country, delivering download speeds of up to 100Mbps.

14.5.24 Figure 14.12 shows the location of the Stratton St Margaret Exchange along Ermin Street approximately 1.5km west of the Site boundary.

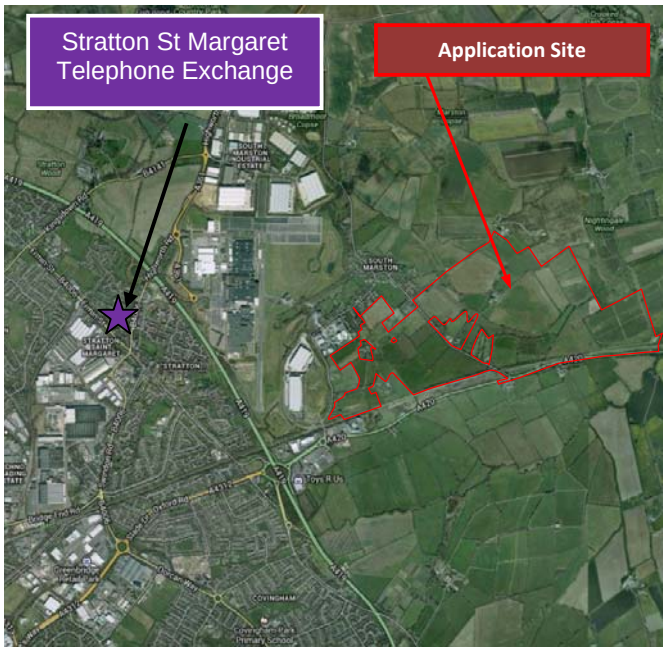

Figure 14.12: Stratton St Margaret Telephone Exchange Location

14.6 MITIGATION MEASURES

Construction stage

14.6.1 In mitigation of the potential construction effects, consideration will be given to the need to shut down supplies while making new connections. Network operators have developed methodologies to permit 'live jointing' or the like whereby the existing network remains fully operational during connection works. During certain operations, and only very occasionally, it remains necessary to temporarily shut down the local network. In such circumstances, the area to be shut down is localised and planned for periods that cause the least disruption. The supplying company is required to give adequate notice to the affected users and ensure that appropriate provision is made for essential supplies.

14.6.2 Potential loss of supply through network damage is mitigated through carefully planning construction activities. The existing and planned networks will be located on the ground and on plans for all contractors to use during implementation. Good working practices, such as 'licence to dig' will be employed, encompassed by the Health & Safety file, to control site operations. Such means of control, which will form part of the proposed CEMP, will substantially reduce the potential risk of damage to the supplying network.

14.6.3 Good working practices and site controls will be maintained throughout the construction stage, again secured through the CEMP, to minimise the risk of network outages to the lowest practical level.

Post-completion stage

14.6.4 In mitigation of the potential post-completion effect of service disruptions to existing users, all service companies have been involved in developing preliminary supply strategies for the Proposed Development. Overall supply capacity and phased load increase assessments have been undertaken from which the supply companies are able to assess the necessary provision. The supply companies are now well advanced with assessments and preliminary proposals to reinforce networks, where necessary, to ensure the supply demands of the Proposed Development and ongoing requirements are met.

14.6.5 The strict regulatory regimes under which all public service supply companies operate dictate that any network expansion results in no loss or reduction of service. These will ensure that the minimum regulatory standards are maintained and that no environmental effect results from supplying the Site with network services.

14.6.6 The service supply companies are developing phased enhancements to their infrastructure, to ensure the availability of capacity and robustness of the network as development phases progress.

14.7 RESIDUAL EFFECTS

Construction stage

14.7.1 Taking into account the mitigation measures outlined, residual effects during the construction stage are assessed as negligible.

Post-completion stage

14.7.2 Again, taking into account the mitigation measures outlined, residual effects following the completion of the Proposed Development are assessed as negligible.

Summary of effects

14.6.5 The effects identified are summarised in Table 14.13 below:

Table 14.13: Summary of effects

Potential effect	Significance (pre-mitigation)	Mitigation measure	Significance of residual effect
Construction stage			
Network outages	**Moderate Adverse**	Care taken when making new connections, live jointing or temporary shutdowns where necessary. Adopting good working practices e.g. 'licence to dig'.	**Negligible**
Post-completion stage			
Shortages of supply	**Minor Adverse**	Development of supply strategies and reinforcements where necessary.	**Negligible**